Ricardo (Zhicong) Xie

Manufacturing / Site Project Intern | CAD-to-Part, Workshop Practice and Coordination

Auckland, New Zealand | 02902729390 | zxie588@aucklanduni.ac.nz | Portfolio: https://zxie588-alt.github.io | GitHub: https://github.com/zxie588-alt

Valid NZ student visa | No sponsorship required | Full-time 17 Nov 2026 - late Feb 2027

PROFILE

- Mechanical engineering master's student with hands-on exposure to laser cutting, CNC machining, 3D printing workflow, automotive structures, CAD modelling and project coordination.
- Interested in manufacturing, production support, site/project engineering and contractor environments where drawing-to-part thinking, practical judgement and workshop communication matter.
- Project screenshots and source links: https://zxie588-alt.github.io

WORK RIGHTS AND AVAILABILITY

- Valid NZ student visa. No sponsorship required.
- Full-time available: 17 Nov 2026 - late Feb 2027 (university break).
- Part-time eligible: up to 25 hours/week during semester. Auckland based; open to relocation for suitable roles.

EDUCATION

University of Auckland - Master of Engineering Studies, Mechanical Engineering | 2026 - Present

Current study and project work in building services, smart manufacturing, CFD/thermal modelling, composite structures and engineering methods.

Relevant coursework: Manufacturing Processes, CNC Machining, Design for Manufacture, Manufacturing and Industrial Systems

Huazhong Agricultural University - Bachelor of Engineering, Mechanical Engineering / Mechatronics | 2021 - 2025

Mechanical design, CAD, electrotechnics, sensors, embedded control and mechatronic system integration.

PROJECT WORK

Manufacturing Workshop Training

Laser cutting, AutoCAD/DXF, CNC machining exposure, 3D printing workflow

- Prepared 2D AutoCAD/DXF geometry for laser cutting and engraving, then compared drawing intent with the physical cut plate.
- Observed and practised CNC machining workflow including workholding, spindle/tool motion, cutting fluid, chip formation and basic workshop safety protocols aligned with general WorkSafe NZ expectations.
- Reviewed machined gear and shaft components to connect CAD features with real manufacturing constraints such as tolerances, edge quality and tool access.

Engineering Practice and Automotive Structure Training

Mechanics testing, robotic arm control, automotive structure walkthrough

- Worked with mechanics testing equipment and interpreted force-displacement results from a physical test setup.
- Practised robotic arm control and basic automation interaction in a lab setting.
- Acted as a small-group lead during automotive structure training, explaining seat/interior structures and vehicle-system context to peers.

Ruggedized Outdoor IoT Sensor Module

SolidWorks, enclosure design, DFM notes, Keil C, concept validation

- Designed a SolidWorks enclosure concept with screw bosses, gasket stack-up, PCB and battery placeholders, cable/strap mounts and service access.
- Applied DFM principles to improve enclosure features for CNC machining discussion and 3D-print prototype planning.
- Added FEA-style screening for corner load, strap lug and screw boss regions; kept an open test list for sealing, impact, thermal exposure and firmware validation.
- Built concept firmware in Keil C with mock sensor readout, battery monitoring and a simple module state machine; documented the electronics and firmware architecture in the portfolio.

NZ Residential MVHR / ERV Ventilation Concept

SolidWorks, AutoCAD/DXF, HVAC sizing, NZBC G4/H1 context, commissioning plan

- Developed a New Zealand residential MVHR/ERV ventilation concept aligned with local building-code requirements.
- Modelled the MVHR unit, supply/extract/outdoor/exhaust ducts, diffusers, grilles and hangers in SolidWorks; exported DXF, STEP, STL, PNG and a component list for review.
- Sized the system for 60 L/s balanced supply/extract in normal operation and 85 L/s boost extract mode, calculating external static pressure targets of about 80 Pa and 120 Pa.
- Checked the design against NZBC G4/H1 and Healthy Homes ventilation context; developed a commissioning test plan covering CO2, humidity, airflow and fan-power measurements.

Orbital AI Data Centre Feasibility Study - Systems Lead

Stakeholders, requirements, ConOps, trade studies, risk, verification

- Led the systems-engineering workstream for a 5-person team concept covering a 24-satellite LEO AI data-centre architecture.
- Set up the stakeholder map, requirements hierarchy, ConOps, architecture options, trade-study criteria, risk register and verification gates.
- Delivered a feasibility-report structure with trade-off matrices, risk logic and verification-gate framework across orbit/coverage, power, thermal rejection, communications, launch mass, cost and sustainability.

TECHNICAL SKILLS

- Proficient in: drawing-to-part thinking, AutoCAD/DXF laser-cut preparation, SolidWorks assemblies, STEP/STL export and component documentation
- Familiar with: CNC machining workflow, workholding, cutting fluid, chip formation, 3D printing workflow and basic workshop safety awareness
- Project support: technician/workshop communication, on-site task support, risk tracking, requirements and verification planning

LEADERSHIP AND COMMUNICATION

- Led team-level integration in the orbital data-centre project and commonly take the organiser role in group work.
- Skilled in technical visual communication and structured documentation; experienced in presenting engineering analysis clearly and producing evidence-based project deliverables.
- Professional working proficiency in English; native Mandarin Chinese.